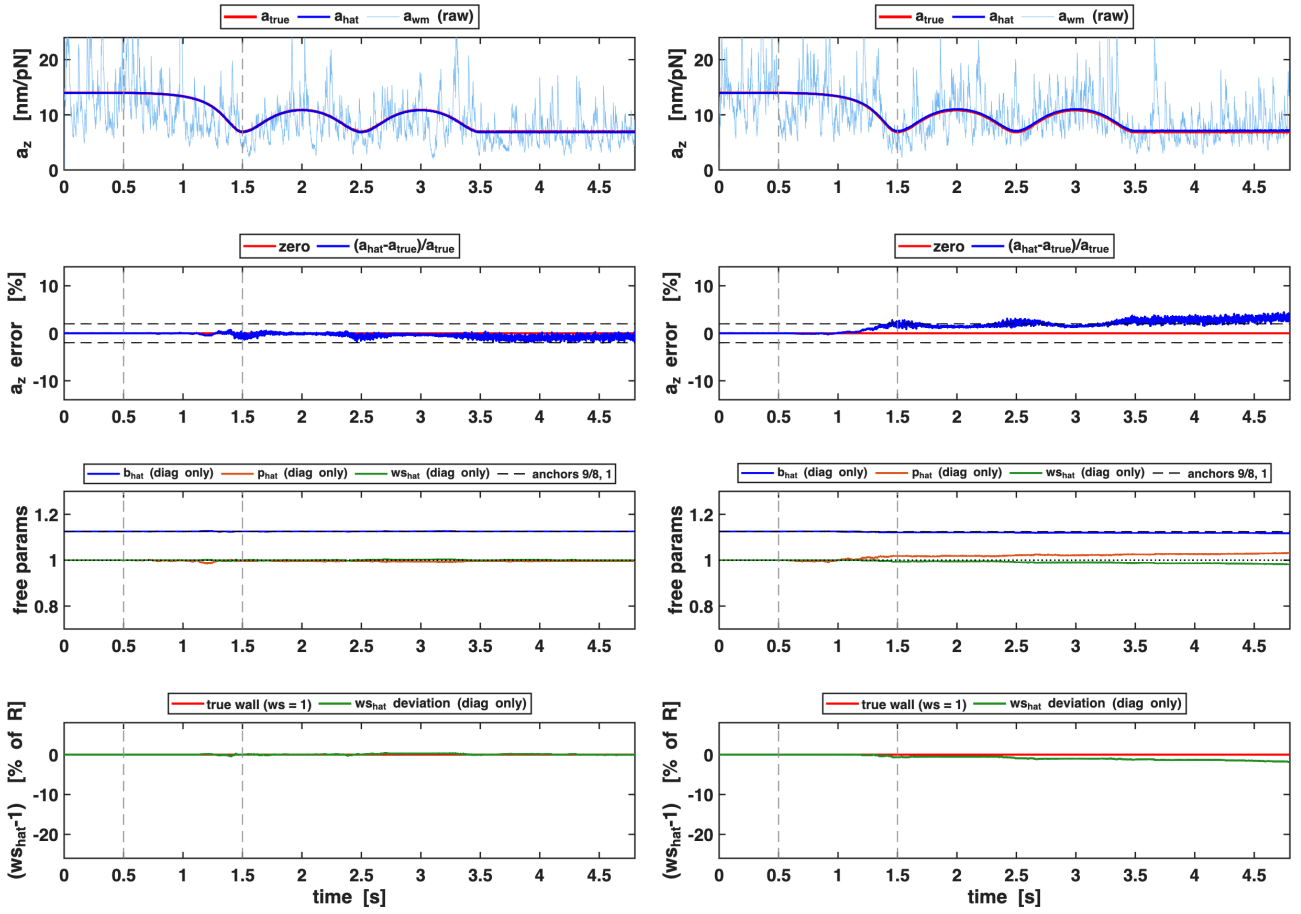
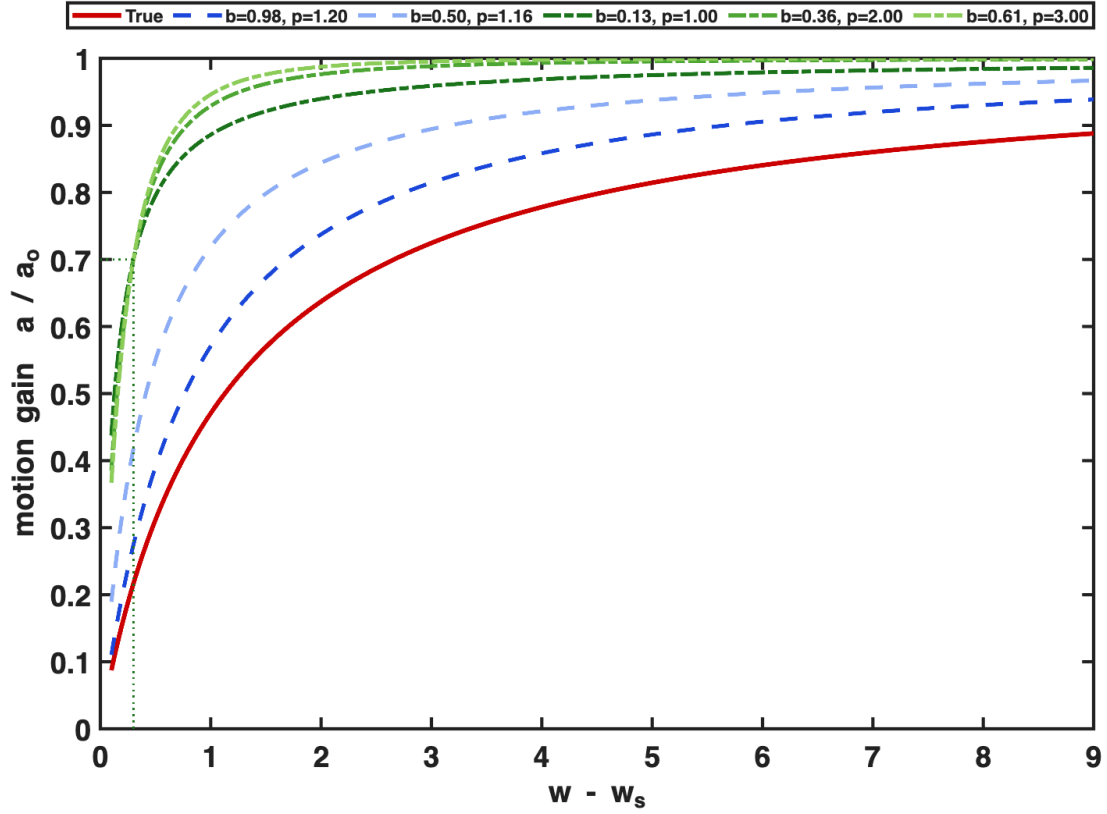


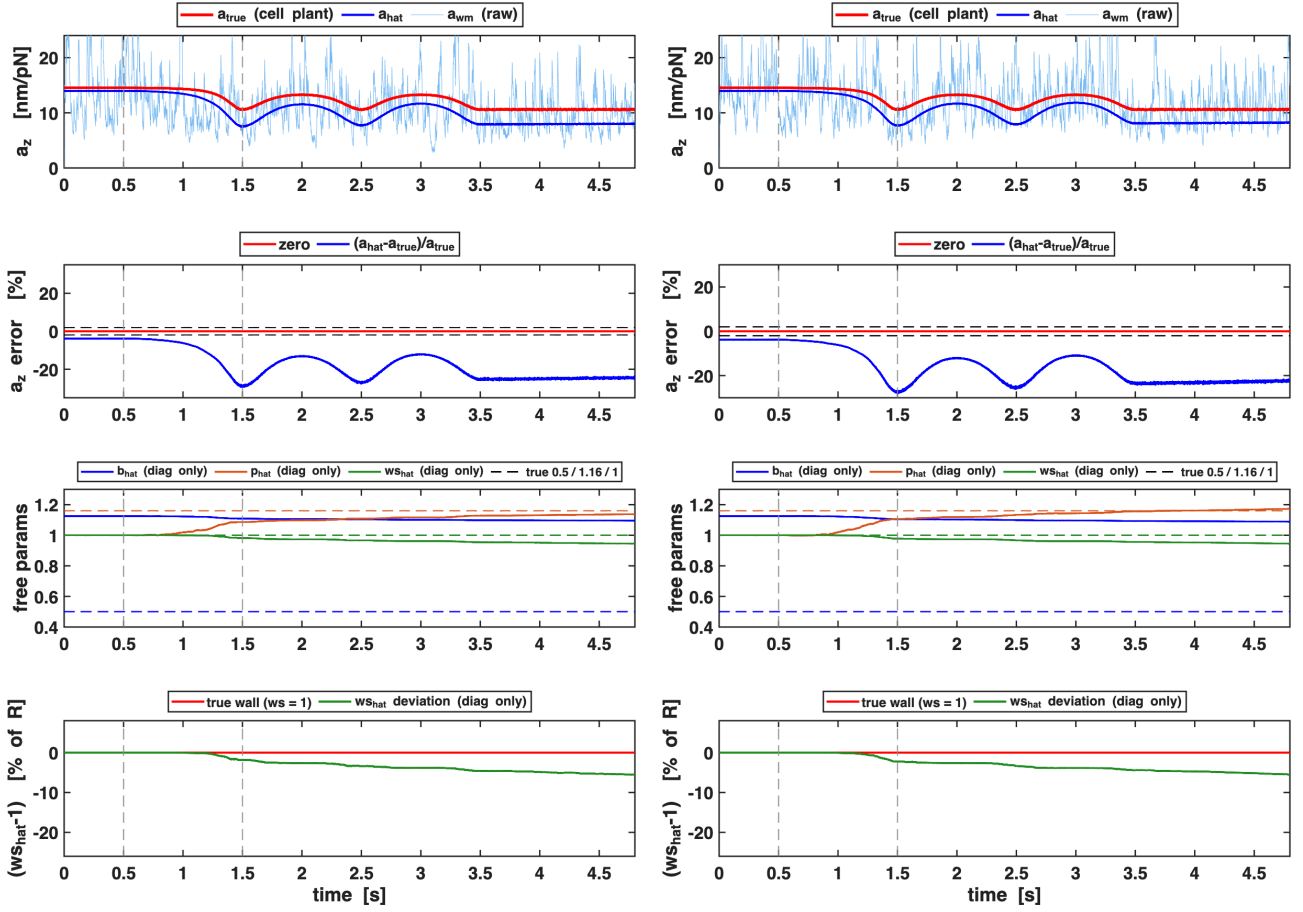
Selected figures (2026-08-03)



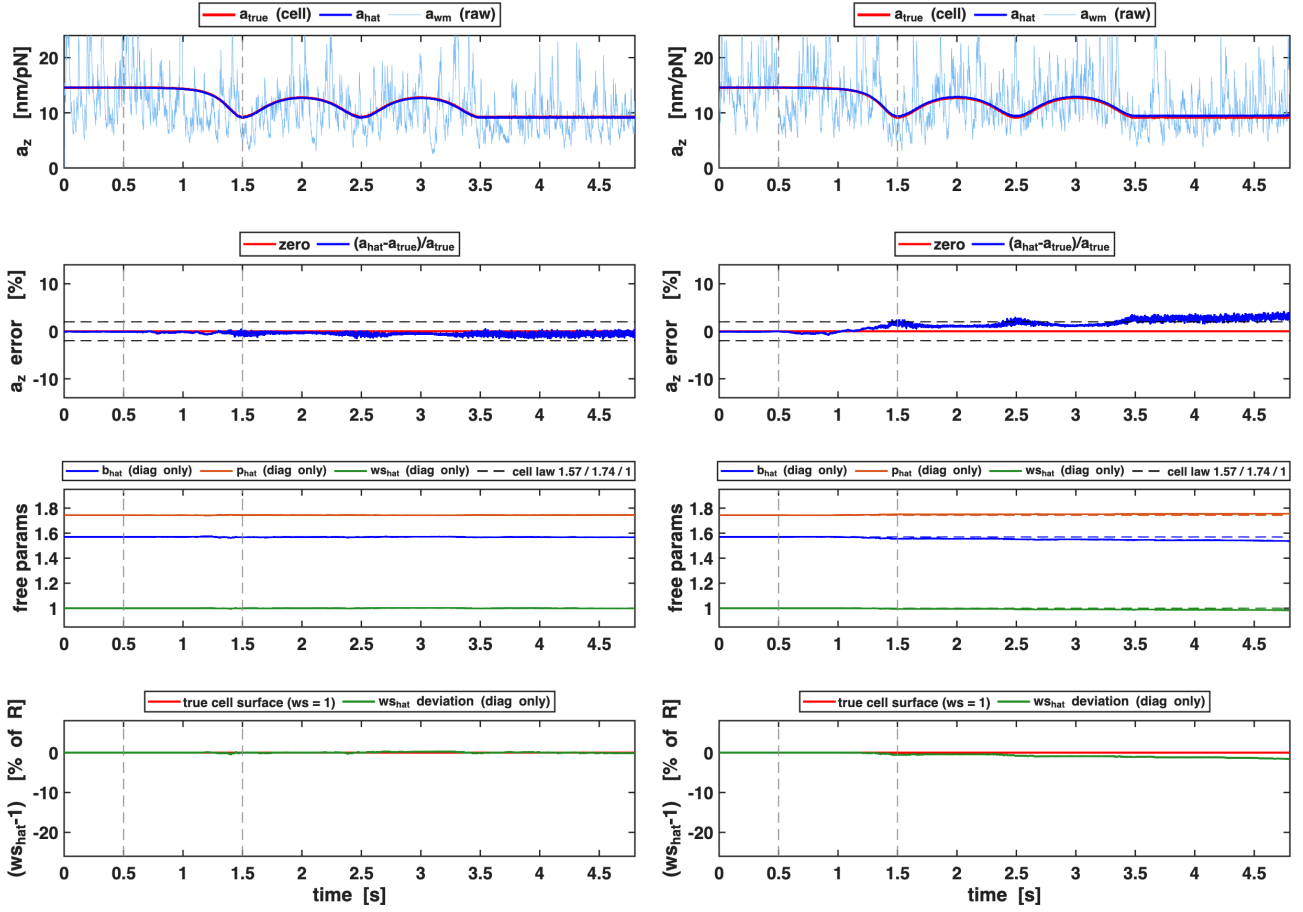
Config 3 — estimate (b, p, w_s) with the calibrated-grade prior $\sqrt{P_{w_s}}[0] = 0.028$ (house 0.111 elsewhere unchanged): seed 7 (left) vs seed 11 (right); the seed-11 $\hat{w}_s$ end deviation shrinks to -1.8% and the residual realization exits through the b, p channels.



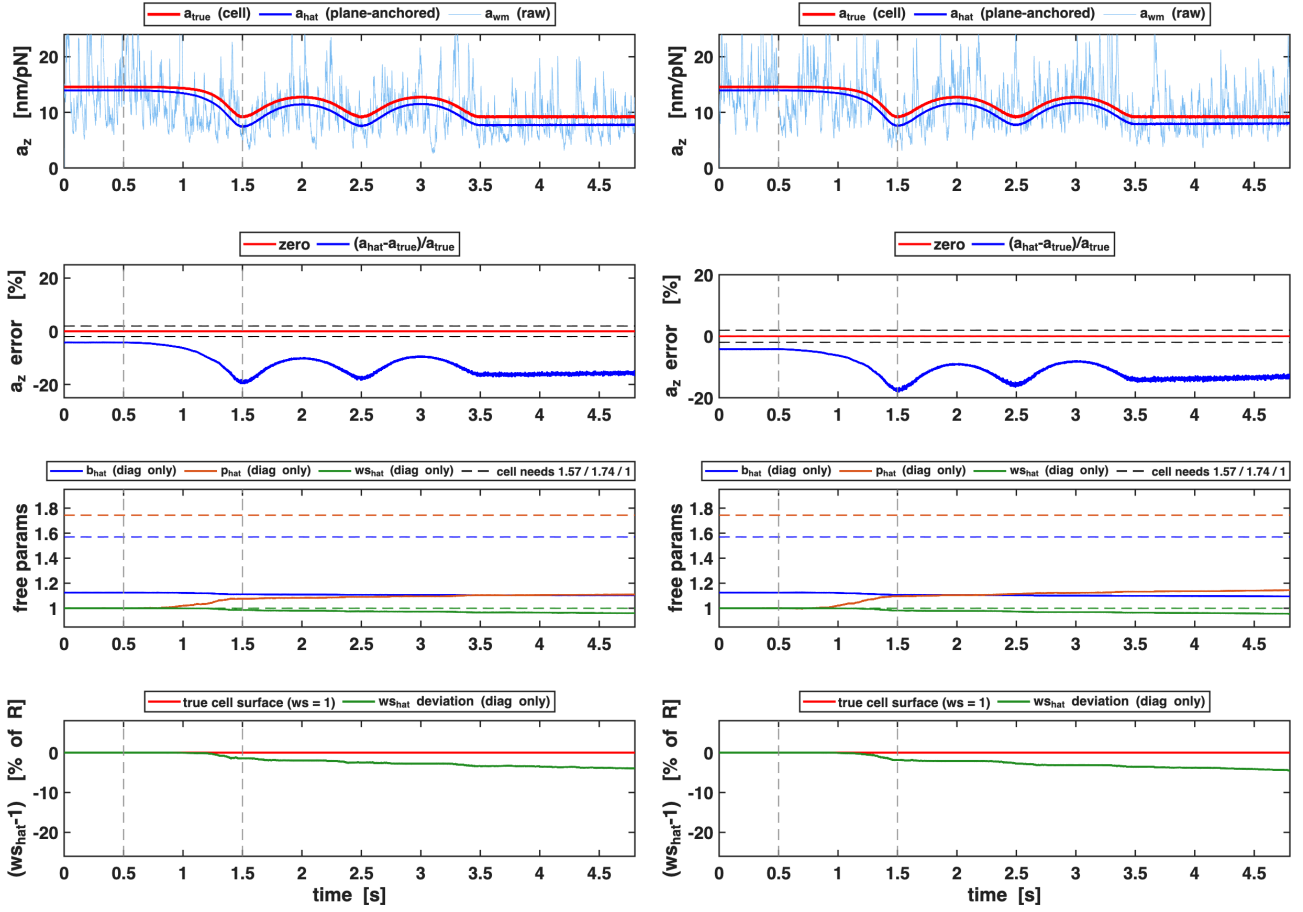
Form B refit to two cell (two-sphere) truths against the rigid-plane truth (red): the family absorbs boundary curvature, but the fitted (b, p) leave the plane anchors $(9/8, 1)$. The three green dash-dot curves are added by hand: all pass through $\bar{a} = 0.70$ at a gap of $0.3 R$ (the rigid plane is at 0.21 there) with the origin still on the surface, and they show the one-parameter freedom a single point leaves — p sets the far-field decay $1 - \bar{a} \sim g^{-p}$ and b follows as $b = rG/(1 - r)$, $r = 0.3^{1/p}$.



Curve-mismatch arm — same config 3 / prior 0.028 setup, but the TRUE z curve is the cell fit $(b, p, w_s) = (0.5, 1.16, 1)$ while the controller keeps every plane-derived prior: gain error blows up (descent peak $\approx 28\%$, hold $\approx -24\%$), the b, p traversal budgets fail by $\sim 50\times$, and $\hat{w}_s$ is dragged -5.5% against its own 2.8% prior — the plane-anchored prior widths cannot absorb a boundary this far outside the family they were derived for.



Cell boundary done properly — plant is the exact two-sphere truth for $R_c = 5 \mu\text{m}$ and the controller carries the cell law $(b, p, w_{s0}) = (1.570, 1.744, 0.809)$ with widths $[0.042, 0.022, 0.015]$, fitted offline to that same published curve (sup 0.086 %): performance returns to the plane baseline (descent 1.10/2.33 %, hold $-0.61/+2.82$ %, $\hat{w}_s$ ending $-0.1/-1.6$ %), so the difficulty was never the curvature but the anchor. This page is the capability bound (“if we knew”), not the operating case.



The operating case — same $R_c = 5 \mu\text{m}$ cell plant, but the controller knows only the plane (anchors 9/8, 1, priors 0.0157/0.0245, origin at the nominal contact): descent peak 19.6/17.8 %, hold $-16.0/ -13.6$ %, b, p stall near their plane anchors (final 1.10/1.11 against the 1.570/1.744 the cell demands), $\hat{w}_s$ absorbs part of the mismatch by drifting $-4.0/ -4.5$ %, and the traversal budgets fail by $25\text{--}40\times$ — a plane-anchored filter meeting a cell is wrong by ~ 15 % and says so, but cannot repair itself.